MEPCO SCHLENK ENGINEERING COLLEGE (AUTONOMOUS), SIVAKASI

Department of Computer Science and Engineering

19CSP01 – Free Open Source Software

Assignment 3: Web Application Development using MySQL (Backend) and PHP (Frontend)

Question

Develop a simple Web application of your choice using:

- MySQL as back-end
- PHP or Python as front-end

Aim

To design and develop a full-stack Student Management System (SMS) web application utilizing MySQL as the relational database backend and PHP as the server-side frontend, incorporating 4 relational tables (departments, courses, students, enrollments) and implementing complete Create, Read, Update, and Delete (CRUD) operations with server-side input validation and dynamic user feedback.

Introduction

Web applications built upon Free and Open Source Software (FOSS) technologies form the foundation of modern internet infrastructure. The LAMP stack (Linux, Apache, MySQL, PHP) represents one of the most widely adopted open-source architectural patterns for dynamic web development.

PHP (Hypertext Preprocessor) is an open-source, server-side scripting language engineered specifically for web development, offering seamless integration with relational databases, dynamic session management, and robust request handling. MySQL is an enterprise-grade, open-source Relational Database Management System (RDBMS) that provides ACID compliance, foreign key relational integrity, and high-performance querying.

In this assignment, a complete Student Management System is constructed. Rather than using an isolated single table, the backend is engineered with 4 interconnected tables structured via Primary Key and Foreign Key constraints, reflecting real-world institutional database architecture.

System Requirements

- Operating System: Windows 11 (64-bit)
- Web Server: Apache HTTP Server 2.4.58 (via XAMPP 8.2.12)
- Database Management System: MySQL / MariaDB 10.4.32 (via phpMyAdmin 5.2.1)
- Server-Side Language: PHP Version 8.2.12
- Frontend Technologies: HTML5, CSS3, JavaScript (Native, Zero External Dependencies)
- Web Browser: Google Chrome / Microsoft Edge

Database Architecture and Relational Schema

The database 'student_db' comprises 4 relational tables establishing 1-to-Many and Many-to-Many relationships:

- departments: Entity table storing department codes, names, and academic buildings (Primary Key: dept_id).
- courses: Entity table maintaining course codes, course titles, and assigned credit units (Primary Key: course_id).
- students: Entity table recording student biodata and linking to departments via a Foreign Key constraint (Primary Key: student_id, Foreign Key: dept_id).
- enrollments: Relational associative table mapping students to their registered courses along with semester and academic grade (Foreign Keys: student_id, course_id).

The detailed schema definition for each table is tabulated below:

Table Name	Field Name	Data Type & Constraints	Role / Description
departments	dept_id	INT AUTO_INCREMENT PRIMARY KEY	Unique Department ID
departments	dept_code	VARCHAR(10) NOT NULL UNIQUE	Department Abbreviation (e.g. CSE)
departments	dept_name	VARCHAR(100) NOT NULL	Full Department Name
departments	building	VARCHAR(50)	Campus Block Location
courses	course_id	INT AUTO_INCREMENT PRIMARY KEY	Unique Course ID
courses	course_code	VARCHAR(15) NOT NULL UNIQUE	Course Code (e.g. 19CSP01)
courses	course_name	VARCHAR(100) NOT NULL	Course Title
courses	credits	INT DEFAULT 3	Academic Credits
students	student_id	INT AUTO_INCREMENT PRIMARY KEY	Unique Student ID
students	name	VARCHAR(100) NOT NULL	Student Full Name

students	roll_no	VARCHAR(20) NOT NULL UNIQUE	University Roll Number
students	email	VARCHAR(150) NOT NULL	Student Email Address
students	phone	VARCHAR(15)	Contact Phone Number
students	dept_id	INT NOT NULL, FOREIGN KEY	FK referencing departments(dept_id)
enrollments	enrollment_id	INT AUTO_INCREMENT PRIMARY KEY	Unique Enrollment ID
enrollments	student_id	INT NOT NULL, FOREIGN KEY	FK referencing students(student_id)
enrollments	course_id	INT NOT NULL, FOREIGN KEY	FK referencing courses(course_id)
enrollments	semester	VARCHAR(10) DEFAULT 'SEM V'	Academic Semester
enrollments	grade	VARCHAR(5) DEFAULT 'A'	Course Performance Grade

Entity-Relationship (ER) Diagram: Chen's Model

The logical structure of the 4-table relational database is formalized using Chen's notation. Rectangles designate Entities (STUDENT, DEPARTMENT, COURSE), diamonds represent Relationships (BELONGS_TO, ENROLLED_IN), and ovals indicate attributes with Primary Keys underlined:

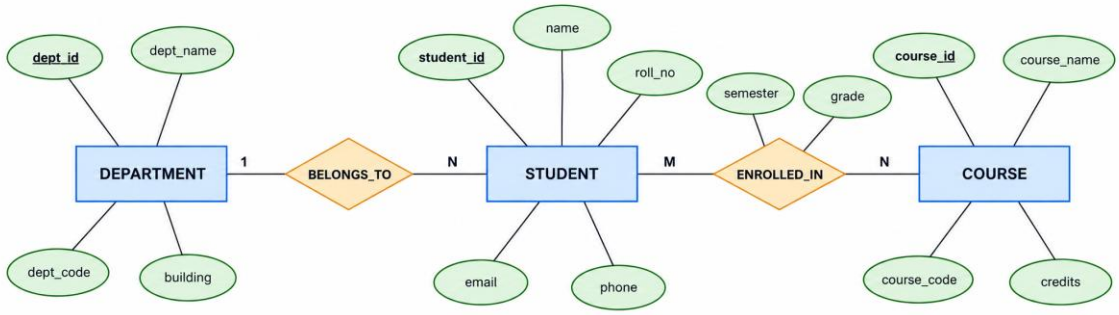


Figure 1: Chen's Model ER Diagram (Entities, Relationships, Attributes, and Cardinalities)

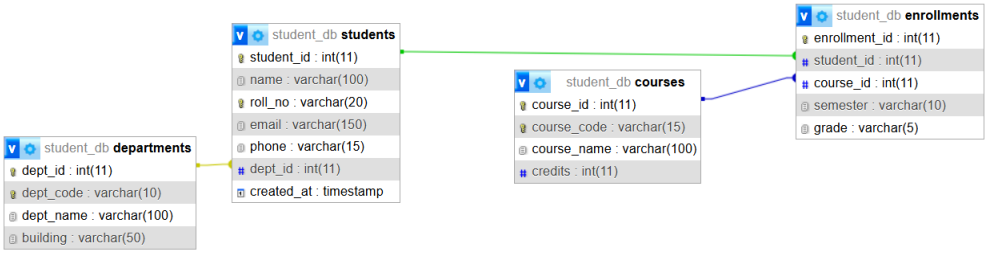


Figure 1b: Relational Schema Architecture Overview

Complete Source Code Implementation

Below are the complete, unmodified source codes for the backend database definitions, database connectivity, and full frontend user interface scripts with embedded PHP server-side logic:

1. Database Schema and Seed Data Script (schema.sql)

```
-- schema.sql: 4 Relational Tables for Student Management System

CREATE DATABASE IF NOT EXISTS student_db;
USE student_db;

-- Table 1: Departments
CREATE TABLE IF NOT EXISTS departments (
    dept_id     INT AUTO_INCREMENT PRIMARY KEY,
    dept_code   VARCHAR(10)  NOT NULL UNIQUE,
    dept_name   VARCHAR(100) NOT NULL,
    building    VARCHAR(50)
);

-- Table 2: Courses
CREATE TABLE IF NOT EXISTS courses (
    course_id   INT AUTO_INCREMENT PRIMARY KEY,
    course_code VARCHAR(15)  NOT NULL UNIQUE,
    course_name VARCHAR(100) NOT NULL,
    credits     INT DEFAULT 3
);

-- Table 3: Students (Foreign Key -> departments)
CREATE TABLE IF NOT EXISTS students (
    student_id  INT AUTO_INCREMENT PRIMARY KEY,
    name        VARCHAR(100) NOT NULL,
    roll_no     VARCHAR(20)  NOT NULL UNIQUE,
    email       VARCHAR(150) NOT NULL,
    phone       VARCHAR(15),
    dept_id     INT NOT NULL,
    created_at  TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (dept_id) REFERENCES departments(dept_id) ON DELETE CASCADE
);

-- Table 4: Enrollments (Relationship Table: connects students & courses)
CREATE TABLE IF NOT EXISTS enrollments (
    enrollment_id INT AUTO_INCREMENT PRIMARY KEY,
    student_id    INT NOT NULL,
    course_id     INT NOT NULL,
    semester      VARCHAR(10) DEFAULT 'SEM V',
    grade         VARCHAR(5)  DEFAULT 'A',
    FOREIGN KEY (student_id) REFERENCES students(student_id) ON DELETE CASCADE,
    FOREIGN KEY (course_id)  REFERENCES courses(course_id)   ON DELETE CASCADE
);
```

```
-- Sample Data Insertion
INSERT INTO departments (dept_code, dept_name, building) VALUES
('CSE', 'Computer Science and Engineering', 'Block A'),
('ECE', 'Electronics and Communication',   'Block B'),
('MECH', 'Mechanical Engineering',         'Block C');

INSERT INTO courses (course_code, course_name, credits) VALUES
('19CSP01', 'Free Open Source Software', 3),
('19CS502', 'Operating Systems',         4),
('19CS503', 'Database Management Systems', 4);

INSERT INTO students (name, roll_no, email, phone, dept_id) VALUES
('Alice Johnson',  '22CS001', 'alice@example.com',  '9876543210', 1),
('Bob Kumar',      '22EC002', 'bob@example.com',    '9876543211', 2),
('Carol Williams', '22ME003', 'carol@example.com',  '9876543212', 3);

INSERT INTO enrollments (student_id, course_id, semester, grade) VALUES
(1, 1, 'SEM V', 'A+'),
(1, 2, 'SEM V', 'A'),
(2, 3, 'SEM V', 'B+'),
(3, 2, 'SEM V', 'A');
```

2. Database Connection Script (db.php)

```
<?php
// db.php – Database connection file
$host     = "localhost";
$user     = "root";
$password = "";         // Change if your MySQL root has a password
$database = "student_db";

$conn = mysqli_connect($host, $user, $password, $database);

if (!$conn) {
   die("<h3 style='color:red;'>Connection Failed: " . mysqli_connect_error() . "</h3>");
}
?>
```

3. Home Dashboard and Read Operation Script (index.php)

```
<?php
// index.php – Display Student Records with Department & Course Relational Data
include 'db.php';

// Handle Delete Operation
if (isset($_GET['delete'])) {
   $id = intval($_GET['delete']);
   mysqli_query($conn, "DELETE FROM students WHERE student_id = $id");
   header("Location: index.php?msg=deleted");
   exit();
}
```

```
// Fetch all students with their Department and Enrolled Courses (JOIN across 4 tables)
$query = "
    SELECT
        s.student_id,
        s.name,
        s.roll_no,
        s.email,
        s.phone,
        d.dept_name,
        d.dept_code,
        GROUP_CONCAT(c.course_code SEPARATOR ', ') AS enrolled_courses
    FROM students s
    JOIN departments d ON s.dept_id = d.dept_id
    LEFT JOIN enrollments e ON s.student_id = e.student_id
    LEFT JOIN courses c ON e.course_id = c.course_id
    GROUP BY s.student_id
    ORDER BY s.student_id DESC
";
$result = mysqli_query($conn, $query);

$msg = '';
if (isset($_GET['msg'])) {
    $map = [
        'added'   => ['Student record created successfully!',   '#d4edda', '#155724'],
        'updated' => ['Student record updated successfully!',   '#d4edda', '#155724'],
        'deleted' => ['Student record deleted successfully!',   '#f8d7da', '#721c24'],
    ];
    if (isset($map[$_GET['msg']])) {
        [$text, $bg, $color] = $map[$_GET['msg']];
        $msg = "<div class='alert' style='background:$bg; color:$color;'>$text</div>";
    }
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>Student Management System - FOSS WebApp</title>
<style>
  * { box-sizing: border-box; margin: 0; padding: 0; }
  body { font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif; background: #f4f7f6; color: #333; }

  header {
    background: #1e3c72;
    background: linear-gradient(to right, #1e3c72, #2a5298);
    color: white;
    padding: 22px 40px;
    display: flex;
    justify-content: space-between;
    align-items: center;
```

```
  box-shadow: 0 4px 10px rgba(0,0,0,0.1);
}
header h1 { font-size: 1.5rem; }
header .subtext { font-size: 0.9rem; opacity: 0.85; }

.container { max-width: 1100px; margin: 30px auto; padding: 0 20px; }

.alert {
  padding: 12px 18px;
  border-radius: 6px;
  margin-bottom: 20px;
  font-weight: 600;
  font-size: 0.95rem;
}

.action-bar {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 20px;
}
.action-bar h2 { font-size: 1.25rem; color: #1e3c72; }

.btn {
  display: inline-block;
  padding: 9px 18px;
  border-radius: 6px;
  font-size: 0.9rem;
  font-weight: 600;
  text-decoration: none;
  border: none;
  cursor: pointer;
  transition: 0.2s ease-in-out;
}
.btn-add { background: #28a745; color: white; }
.btn-add:hover { background: #218838; }
.btn-edit { background: #ffc107; color: #212529; margin-right: 6px; }
.btn-edit:hover { background: #e0a800; }
.btn-delete { background: #dc3545; color: white; }
.btn-delete:hover { background: #c82333; }

table {
  width: 100%;
  border-collapse: collapse;
  background: white;
  border-radius: 8px;
  overflow: hidden;
  box-shadow: 0 2px 10px rgba(0,0,0,0.06);
}
thead { background: #1e3c72; color: white; }
thead th { padding: 14px 16px; text-align: left; font-size: 0.9rem; letter-spacing: 0.5px; }
```

```
 tbody tr { border-bottom: 1px solid #eee; transition: background 0.15s; }
 tbody tr:hover { background: #f9fbfd; }
 tbody td { padding: 12px 16px; font-size: 0.9rem; }

 .badge {
  background: #e7f1ff;
  color: #0d6efd;
  padding: 4px 10px;
  border-radius: 12px;
  font-size: 0.8rem;
  font-weight: 600;
 }
 .badge-courses {
  background: #e8f5e9;
  color: #2e7d32;
  padding: 4px 10px;
  border-radius: 12px;
  font-size: 0.8rem;
  font-weight: 600;
 }
 .empty-row { text-align: center; padding: 40px; color: #888; font-style: italic; }
</style>
</head>
<body>

<header>
 <div>
  <h1> 🎓 Student Management System</h1>
  <span class="subtext">FOSS Web Application (PHP & MySQL Relational CRUD)</span>
 </div>
 <div class="subtext">4 Relational Tables</div>
</header>

<div class="container">
 <?= $msg ?>

 <div class="action-bar">
  <h2>Student Records List (<?= mysqli_num_rows($result) ?> Registered)</h2>
  <!-- Combined CRUD: CREATE BUTTON -->
  <a href="add.php" class="btn btn-add">+ Add New Student</a>
 </div>

 <table>
  <thead>
   <tr>
    <th>Roll No</th>
    <th>Student Name</th>
    <th>Department</th>
    <th>Enrolled Courses</th>
    <th>Email</th>
    <th>Phone</th>
```

```
      <th>Actions</th>
    </tr>
  </thead>
  <tbody>
    <?php if (mysqli_num_rows($result) === 0): ?>
      <tr><td colspan="7" class="empty-row">No student records found. Click "+ Add New Student" to insert one.</td></tr>
    <?php else: ?>
      <?php while ($row = mysqli_fetch_assoc($result)): ?>
      <tr>
        <td><span class="badge"><?= htmlspecialchars($row['roll_no']) ?></span></td>
        <td><strong><?= htmlspecialchars($row['name']) ?></strong></td>
        <td><?= htmlspecialchars($row['dept_name']) ?> (<?= htmlspecialchars($row['dept_code']) ?>)</td>
        <td>
          <span class="badge-courses"><?= htmlspecialchars($row['enrolled_courses'] ?: 'None') ?></span>
        </td>
        <td><?= htmlspecialchars($row['email']) ?></td>
        <td><?= htmlspecialchars($row['phone'] ?: '—') ?></td>
        <td>
          <!-- Combined CRUD: UPDATE & DELETE BUTTONS -->
          <a href="edit.php?id=<?= $row['student_id'] ?>" class="btn btn-edit">Edit</a>
          <a href="index.php?delete=<?= $row['student_id'] ?>"
            class="btn btn-delete"
            onclick="return confirm('Are you sure you want to delete this student record?');">Delete</a>
        </td>
      </tr>
      <?php endwhile; ?>
    <?php endif; ?>
  </tbody>
 </table>
</div>

</body>
</html>
```

4. Student Creation Form and Validation Script (add.php)

```
<?php
// add.php – Create New Student with Department and Course Enrollment
include 'db.php';

// Fetch departments and courses from database
$depts_res   = mysqli_query($conn, "SELECT * FROM departments ORDER BY dept_name");
$courses_res = mysqli_query($conn, "SELECT * FROM courses ORDER BY course_name");

$errors = [];

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
  $name      = trim($_POST['name'] ?? '');
  $roll_no   = trim($_POST['roll_no'] ?? '');
  $email     = trim($_POST['email'] ?? '');
```

```
    $phone     = trim($_POST['phone'] ?? '');
    $dept_id   = intval($_POST['dept_id'] ?? 0);
    $course_id = intval($_POST['course_id'] ?? 0);

    // Validation
    if ($name === '') $errors[] = "Student Name is required.";
    if ($roll_no === '') $errors[] = "Roll Number is required.";
    if ($dept_id <= 0) $errors[] = "Please select a Department.";
    if ($course_id <= 0) $errors[] = "Please select a Course to enroll.";
    if ($email === '') {
        $errors[] = "Email is required.";
    } elseif (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
        $errors[] = "Invalid Email address format.";
    }

    if (empty($errors)) {
        $name_esc    = mysqli_real_escape_string($conn, $name);
        $roll_esc    = mysqli_real_escape_string($conn, $roll_no);
        $email_esc   = mysqli_real_escape_string($conn, $email);
        $phone_esc   = mysqli_real_escape_string($conn, $phone);

        // Insert into students table
        $sql_student = "INSERT INTO students (name, roll_no, email, phone, dept_id)
                        VALUES ('$name_esc', '$roll_esc', '$email_esc', '$phone_esc', $dept_id)";

        if (mysqli_query($conn, $sql_student)) {
            $new_student_id = mysqli_insert_id($conn);
            // Insert into enrollments table (relationship)
            $sql_enroll = "INSERT INTO enrollments (student_id, course_id, semester, grade)
                           VALUES ($new_student_id, $course_id, 'SEM V', 'A')";
            mysqli_query($conn, $sql_enroll);

            header("Location: index.php?msg=added");
            exit();
        } else {
            $errors[] = "Database Error: " . mysqli_error($conn);
        }
    }
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>Add Student - FOSS WebApp</title>
<style>
  * { box-sizing: border-box; margin: 0; padding: 0; }
  body { font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif; background: #f4f7f6; color: #333; }

  header {
```

```
  background: linear-gradient(to right, #1e3c72, #2a5298);
  color: white;
  padding: 22px 40px;
}
header h1 { font-size: 1.5rem; }

.card {
  max-width: 600px;
  margin: 35px auto;
  background: white;
  padding: 30px;
  border-radius: 8px;
  box-shadow: 0 2px 12px rgba(0,0,0,0.08);
}
.card h2 { color: #1e3c72; margin-bottom: 20px; font-size: 1.3rem; }

.form-group { margin-bottom: 16px; }
label { display: block; font-size: 0.88rem; font-weight: 600; margin-bottom: 6px; color: #555; }
input[type="text"], input[type="email"], select {
  width: 100%;
  padding: 10px 14px;
  border: 1px solid #ced4da;
  border-radius: 5px;
  font-size: 0.95rem;
}
input:focus, select:focus {
  outline: none;
  border-color: #2a5298;
  box-shadow: 0 0 0 3px rgba(42,82,152,0.15);
}

.btn {
  display: inline-block;
  padding: 10px 20px;
  border-radius: 5px;
  font-size: 0.95rem;
  font-weight: 600;
  border: none;
  cursor: pointer;
  text-decoration: none;
}
.btn-submit { background: #28a745; color: white; margin-right: 10px; }
.btn-submit:hover { background: #218838; }
.btn-back { background: #6c757d; color: white; }
.btn-back:hover { background: #5a6268; }

.error-box {
  background: #f8d7da;
  color: #721c24;
  padding: 12px 16px;
  border-radius: 5px;
```

```
   margin-bottom: 18px;
   font-size: 0.9rem;
  }
  .error-box ul { margin-left: 20px; }
</style>
</head>
<body>

<header>
  <h1> 🎓 Student Management System</h1>
</header>

<div class="card">
  <h2> ➕ Create New Student Record</h2>

  <?php if (!empty($errors)): ?>
  <div class="error-box">
    <strong>Validation Error:</strong>
    <ul>
      <?php foreach ($errors as $err): ?>
        <li><?= htmlspecialchars($err) ?></li>
      <?php endforeach; ?>
    </ul>
  </div>
  <?php endif; ?>

  <form method="POST" action="add.php">
    <div class="form-group">
      <label>Full Student Name *</label>
      <input type="text" name="name" placeholder="e.g. David Miller" value="<?=
htmlspecialchars($_POST['name'] ?? '') ?>" required>
    </div>

    <div class="form-group">
      <label>Roll Number *</label>
      <input type="text" name="roll_no" placeholder="e.g. 22CS004" value="<?=
htmlspecialchars($_POST['roll_no'] ?? '') ?>" required>
    </div>

    <div class="form-group">
      <label>Department (Relational Table 1) *</label>
      <select name="dept_id" required>
        <option value="">-- Choose Department --</option>
        <?php while ($d = mysqli_fetch_assoc($depts_res)): ?>
          <option value="<?= $d['dept_id'] ?>" <?= (isset($_POST['dept_id']) && $_POST['dept_id'] ==
$d['dept_id']) ? 'selected' : '' ?>>
            <?= htmlspecialchars($d['dept_name']) ?> (<?= htmlspecialchars($d['dept_code']) ?>)
          </option>
        <?php endwhile; ?>
      </select>
    </div>
```

```
    <div class="form-group">
      <label>Course Enrollment (Relational Table 2 & 4) *</label>
      <select name="course_id" required>
        <option value="">-- Choose Course to Enroll --</option>
        <?php while ($c = mysqli_fetch_assoc($courses_res)): ?>
          <option value="<?= $c['course_id'] ?>" <?= (isset($_POST['course_id']) && $_POST['course_id']
== $c['course_id']) ? 'selected' : '' ?>>
            <?= htmlspecialchars($c['course_name']) ?> (<?= htmlspecialchars($c['course_code']) ?>)
          </option>
        <?php endwhile; ?>
      </select>
    </div>

    <div class="form-group">
      <label>Email Address *</label>
      <input type="email" name="email" placeholder="e.g. david@example.com" value="<?=
htmlspecialchars($_POST['email'] ?? '') ?>" required>
    </div>

    <div class="form-group">
      <label>Phone Number</label>
      <input type="text" name="phone" placeholder="e.g. 9876543215" value="<?=
htmlspecialchars($_POST['phone'] ?? '') ?>">
    </div>

    <button type="submit" class="btn btn-submit">Save Student Record</button>
    <a href="index.php" class="btn btn-back">Cancel</a>
  </form>
</div>

</body>
</html>
```

5. Student Update Form and Processing Script (edit.php)

```
<?php
// edit.php – Update Existing Student Record
include 'db.php';

$id = intval($_GET['id'] ?? 0);
if ($id <= 0) {
    header("Location: index.php");
    exit();
}

// Fetch existing student details
$res = mysqli_query($conn, "SELECT * FROM students WHERE student_id = $id");
if (mysqli_num_rows($res) === 0) {
    header("Location: index.php");
    exit();
}
```

```php
$student = mysqli_fetch_assoc($res);

// Fetch departments for dropdown
$depts_res = mysqli_query($conn, "SELECT * FROM departments ORDER BY dept_name");

$errors = [];

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $name    = trim($_POST['name'] ?? '');
    $roll_no = trim($_POST['roll_no'] ?? '');
    $email   = trim($_POST['email'] ?? '');
    $phone   = trim($_POST['phone'] ?? '');
    $dept_id = intval($_POST['dept_id'] ?? 0);

    // Validation
    if ($name === '') $errors[] = "Student Name is required.";
    if ($roll_no === '') $errors[] = "Roll Number is required.";
    if ($dept_id <= 0) $errors[] = "Please select a Department.";
    if ($email === '') {
        $errors[] = "Email is required.";
    } elseif (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
        $errors[] = "Invalid Email address format.";
    }

    if (empty($errors)) {
        $name_esc  = mysqli_real_escape_string($conn, $name);
        $roll_esc  = mysqli_real_escape_string($conn, $roll_no);
        $email_esc = mysqli_real_escape_string($conn, $email);
        $phone_esc = mysqli_real_escape_string($conn, $phone);

        $sql_update = "UPDATE students SET
                    name = '$name_esc',
                    roll_no = '$roll_esc',
                    dept_id = $dept_id,
                    email = '$email_esc',
                    phone = '$phone_esc'
                   WHERE student_id = $id";

        if (mysqli_query($conn, $sql_update)) {
            header("Location: index.php?msg=updated");
            exit();
        } else {
            $errors[] = "Database Error: " . mysqli_error($conn);
        }
    }
    // Update local variable on error
    $student['name']    = $name;
    $student['roll_no'] = $roll_no;
    $student['email']   = $email;
    $student['phone']   = $phone;
    $student['dept_id'] = $dept_id;
```

```
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>Edit Student - FOSS WebApp</title>
<style>
 * { box-sizing: border-box; margin: 0; padding: 0; }
 body { font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif; background: #f4f7f6; color: #333; }

 header {
  background: linear-gradient(to right, #1e3c72, #2a5298);
  color: white;
  padding: 22px 40px;
 }
 header h1 { font-size: 1.5rem; }

 .card {
  max-width: 600px;
  margin: 35px auto;
  background: white;
  padding: 30px;
  border-radius: 8px;
  box-shadow: 0 2px 12px rgba(0,0,0,0.08);
 }
 .card h2 { color: #e65100; margin-bottom: 20px; font-size: 1.3rem; }

 .form-group { margin-bottom: 16px; }
 label { display: block; font-size: 0.88rem; font-weight: 600; margin-bottom: 6px; color: #555; }
 input[type="text"], input[type="email"], select {
  width: 100%;
  padding: 10px 14px;
  border: 1px solid #ced4da;
  border-radius: 5px;
  font-size: 0.95rem;
 }
 input:focus, select:focus {
  outline: none;
  border-color: #f9a825;
  box-shadow: 0 0 0 3px rgba(249,168,37,0.18);
 }

 .btn {
  display: inline-block;
  padding: 10px 20px;
  border-radius: 5px;
  font-size: 0.95rem;
  font-weight: 600;
  border: none;
```

```
  cursor: pointer;
  text-decoration: none;
 }
 .btn-update { background: #ffc107; color: #212529; margin-right: 10px; }
 .btn-update:hover { background: #e0a800; }
 .btn-back { background: #6c757d; color: white; }
 .btn-back:hover { background: #5a6268; }

 .error-box {
  background: #f8d7da;
  color: #721c24;
  padding: 12px 16px;
  border-radius: 5px;
  margin-bottom: 18px;
  font-size: 0.9rem;
 }
 .error-box ul { margin-left: 20px; }
</style>
</head>
<body>

<header>
 <h1> 🎓 Student Management System</h1>
</header>

<div class="card">
 <h2> 🖊 Edit / Update Student Record (ID: #<?= $id ?>)</h2>

 <?php if (!empty($errors)): ?>
 <div class="error-box">
  <strong>Validation Error:</strong>
  <ul>
   <?php foreach ($errors as $err): ?>
    <li><?= htmlspecialchars($err) ?></li>
   <?php endforeach; ?>
  </ul>
 </div>
 <?php endif; ?>

 <form method="POST" action="edit.php?id=<?= $id ?>">
  <div class="form-group">
   <label>Full Student Name *</label>
   <input type="text" name="name" value="<?= htmlspecialchars($student['name']) ?>" required>
  </div>

  <div class="form-group">
   <label>Roll Number *</label>
   <input type="text" name="roll_no" value="<?= htmlspecialchars($student['roll_no']) ?>" required>
  </div>
```

```
    <div class="form-group">
      <label>Department (Relational Table 1) *</label>
      <select name="dept_id" required>
        <?php while ($d = mysqli_fetch_assoc($depts_res)): ?>
          <option value="<?= $d['dept_id'] ?>" <?= ($student['dept_id'] == $d['dept_id']) ? 'selected' : '' ?>>
            <?= htmlspecialchars($d['dept_name']) ?> (<?= htmlspecialchars($d['dept_code']) ?>)
          </option>
        <?php endwhile; ?>
      </select>
    </div>

    <div class="form-group">
      <label>Email Address *</label>
      <input type="email" name="email" value="<?= htmlspecialchars($student['email']) ?>" required>
    </div>

    <div class="form-group">
      <label>Phone Number</label>
      <input type="text" name="phone" value="<?= htmlspecialchars($student['phone']) ?>">
    </div>

    <button type="submit" class="btn btn-update">Update Student Record</button>
    <a href="index.php" class="btn btn-back">Cancel</a>
  </form>
</div>

</body>
</html>
```

Experimental Verification and Results

The full CRUD lifecycle of the web application was verified locally using the Apache server and MySQL daemon:

A. Read Operation (Dashboard with Combined CRUD Actions)

Navigating to http://localhost/webapp/index.php displays the styled student directory with real-time relational join data and combined CRUD action buttons:

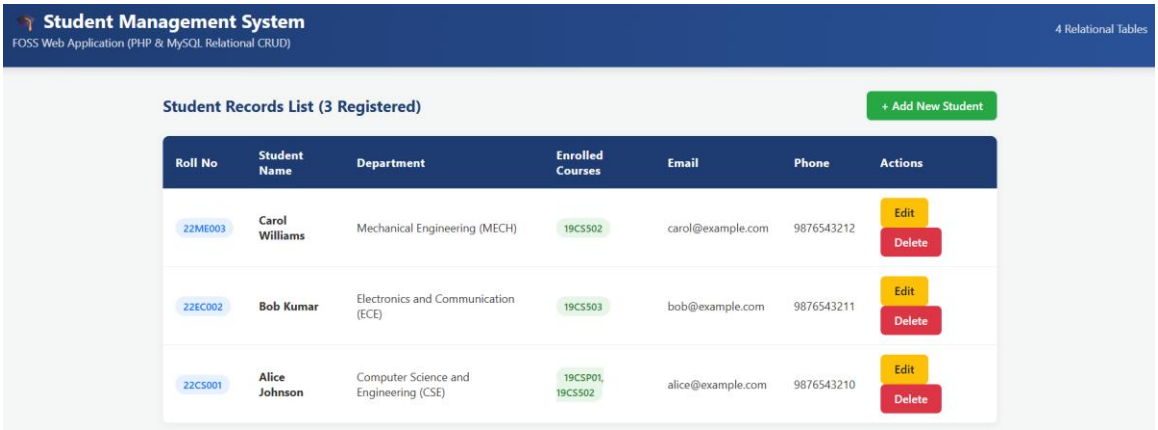


Figure 2: Home Page (Read Operation) Displaying Relational Data and Combined CRUD Action Buttons

B. Create Operation (Add Student Form and Success Notification)

Clicking the '+ Add New Student' button renders the registration form with dynamic department and course selection dropdowns. Submitting valid data inserts the record and renders a green confirmation banner:

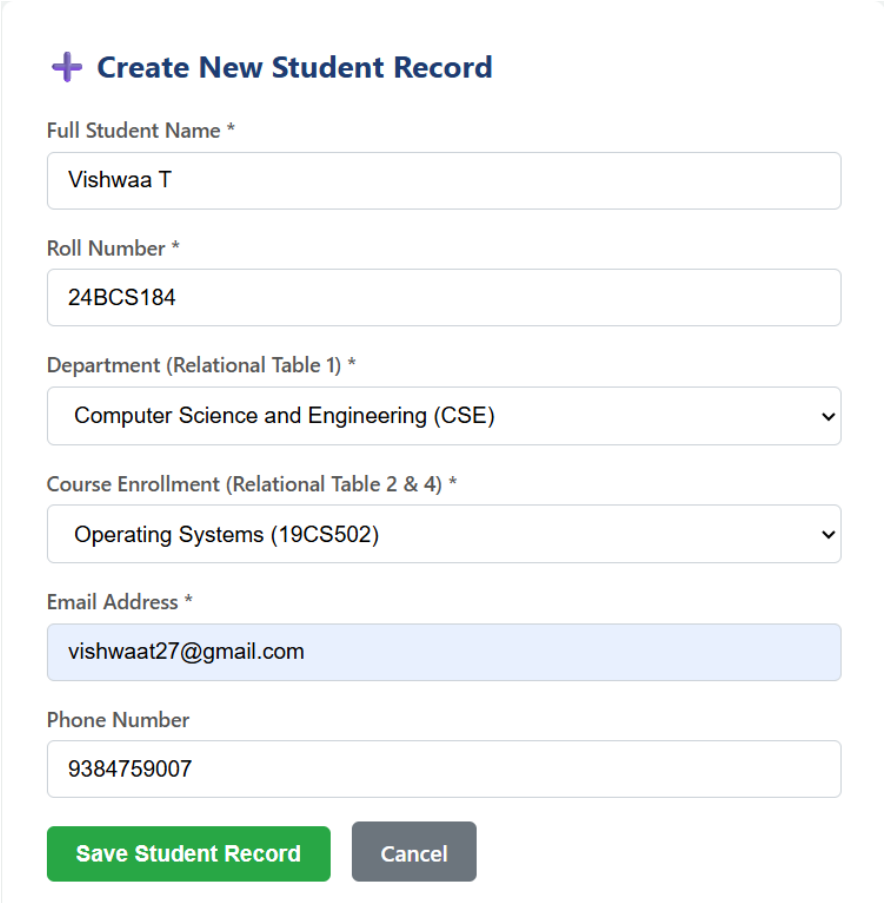


Figure 3a: Create Operation – Add Student Form with Dynamic Relational Dropdowns

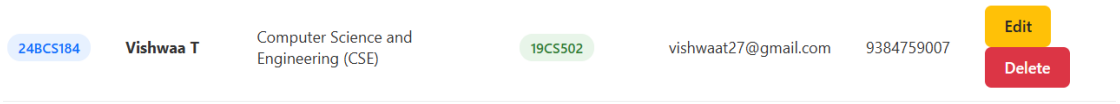


Figure 3b: Create Operation – Successful Record Creation with Green Status Notification Banner

C. Update Operation (Edit Student Record)

Clicking 'Edit' on a student record opens the pre-populated update form. Modifying attributes and clicking 'Update Student Record' saves the changes and displays an updated status alert:

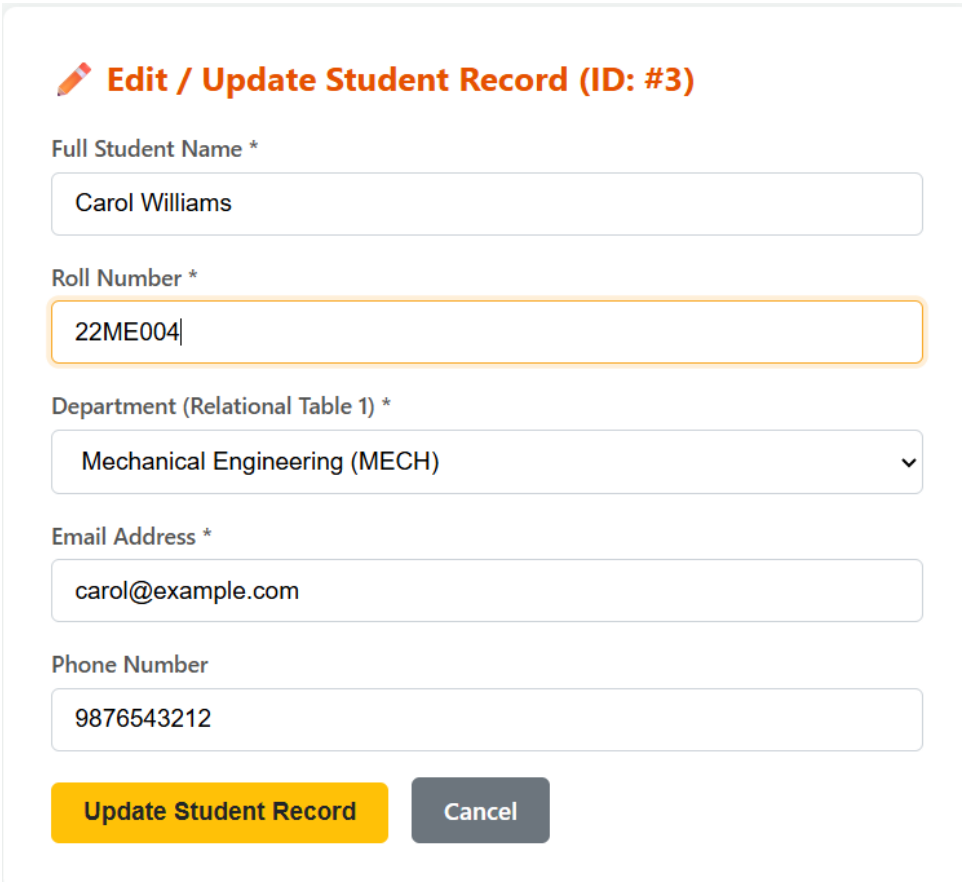


Figure 4a: Update Operation – Pre-populated Student Modification Form

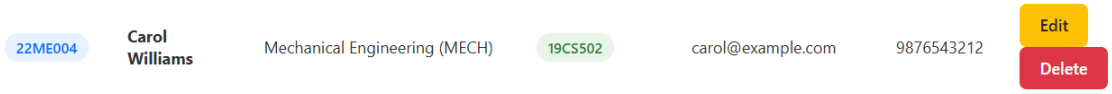


Figure 4b: Update Operation – Successful Record Update with Status Notification Banner

D. Delete Operation (Student Record Removal)

Clicking the 'Delete' button triggers a JavaScript confirmation modal. Confirming execution removes the record from MySQL and immediately refreshes the dashboard with a red alert banner:

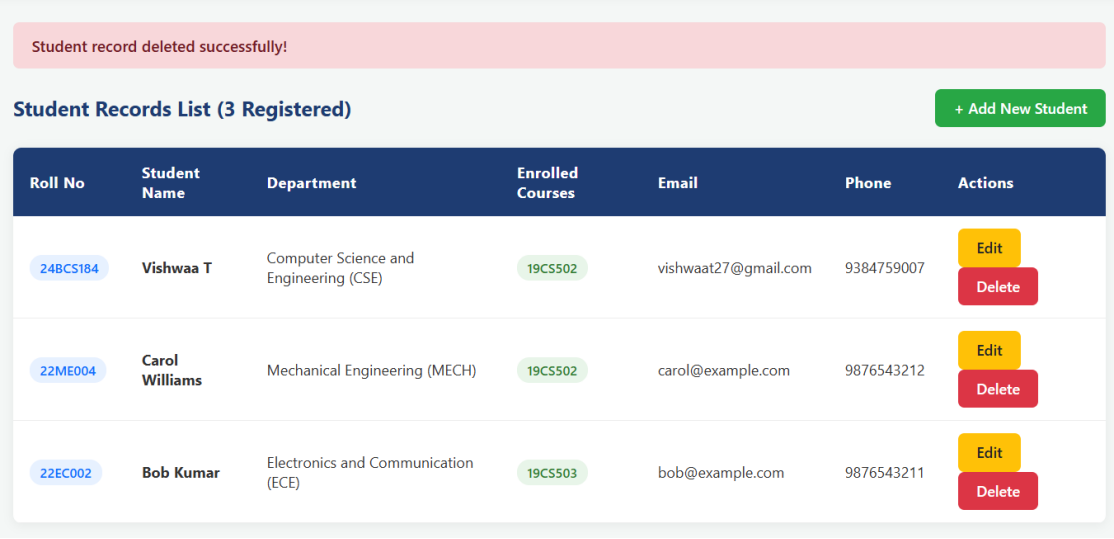


Figure 5: Delete Operation – Instant Red Alert Notification Banner Confirming Record Deletion

Summary of SQL CRUD Queries Executed

CREATE Query:

INSERT INTO students (name, roll_no, email, phone, dept_id) VALUES ('David Miller', '22CS004', 'david@example.com', '9876543213', 1);

READ Query:

SELECT s., d.dept_name, GROUP_CONCAT(c.course_code) FROM students s JOIN departments d ON s.dept_id=d.dept_id LEFT JOIN enrollments e ON s.student_id=e.student_id LEFT JOIN courses c ON e.course_id=c.course_id GROUP BY s.student_id;*

UPDATE Query:

UPDATE students SET name='David Miller', phone='9876543299', dept_id=2 WHERE student_id=4;

DELETE Query:

DELETE FROM students WHERE student_id=4;

Output / Result

A dynamic, full-stack Student Management System web application was successfully designed and deployed using MySQL as the relational backend and PHP as the frontend. The system incorporates 4 relational tables (departments, courses, students, enrollments) linked via foreign keys. All four CRUD operations (Create, Read, Update, Delete) were comprehensively executed with responsive user feedback banners and strict server-side validation. All experimental workflows were verified on the local Apache and MySQL runtime.

Conclusion

This assignment provided comprehensive practical experience in developing full-stack database-driven web applications using foundational FOSS technologies. The synergy between PHP's server-side processing and MySQL's relational data integrity was demonstrated through multi-table relational schema architecture, input sanitization, dynamic query construction, and responsive user experience design. The project underscores the effectiveness of open-source tooling in engineering scalable, enterprise-grade information systems.